

2014 HKDSE Physics

Paper 2 Elective Sections

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Section A Astronomy and Space Science

Question 1.1

Mercury is 0.39 AU from the Sun. Which of the following is **NOT** a possible distance between Mercury and the Earth? Assume that the orbits of Mercury and the Earth are circular and co-planar.

- A. 1.20 AU
- B. 1.00 AU
- C. 0.78 AU
- D. 0.50 AU

Official answer: D

D. The Earth has orbital radius 1 AU. The separation ranges from $1 - 0.39 = 0.61$ AU to $1 + 0.39 = 1.39$ AU. Only 0.50 AU is outside this interval.

Question 1.2

Given that a typical galaxy in the form of a circular disc is of diameter 10^5 ly and thickness 10^3 ly containing about 10^{11} stars, estimate the average separation between two neighbouring stars within the galaxy assuming that the stars are uniformly distributed.

- A. 4.3 ly
- B. 6.8 ly
- C. 8.9 ly
- D. 43 ly

Official answer: A

A. Volume per star = $\pi(5 \times 10^4)^2(10^3)/10^{11} = 78.54$ ly³. The characteristic spacing is its cube root, 4.28 ly ≈ 4.3 ly. Use a volume, not just the disc area.

Question 1.3

If the Sun is on the ecliptic as shown in Figure (1), which of the following positions of the Sun in Figure (2) shows the view seen by an observer at a latitude 23.5° north of the equator at noon?

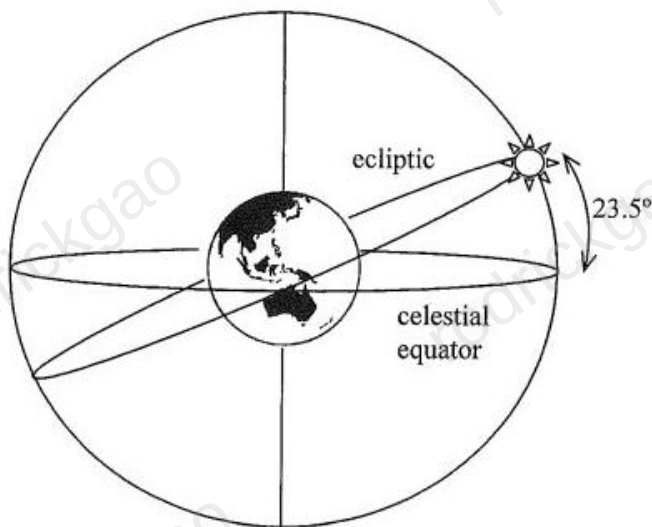


Figure (1)

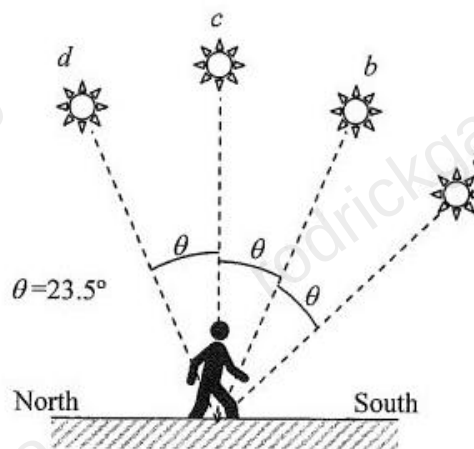


Figure (2)

- A. Position *a*
- B. Position *b*
- C. Position *c*
- D. Position *d*

Official answer: C

C. The Sun has declination $+23.5^\circ$. At latitude $+23.5^\circ$, its noon altitude is $90^\circ - |23.5^\circ - 23.5^\circ| = 90^\circ$, directly overhead at *c*.

Question 1.4

The violet line (410 nm) of the hydrogen spectrum from a distant celestial body is blue shifted and its wavelength appears 50 nm shorter when observed. What is the observed wavelength of the red line (656 nm) from the same source?

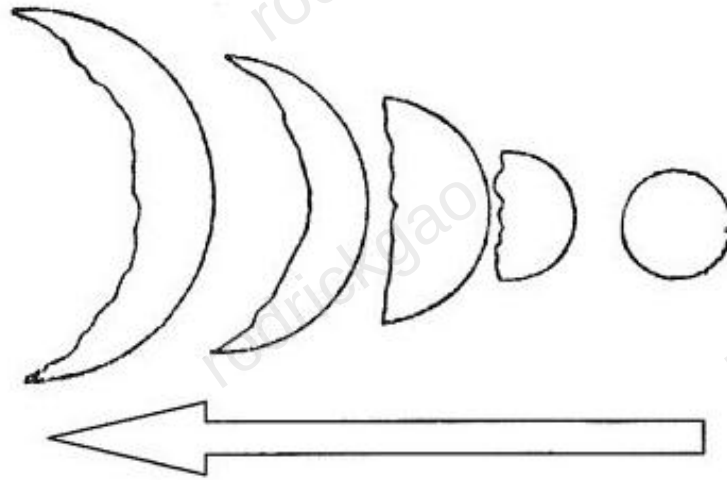
- A. 576 nm
- B. 606 nm
- C. 706 nm
- D. 736 nm

Official answer: A

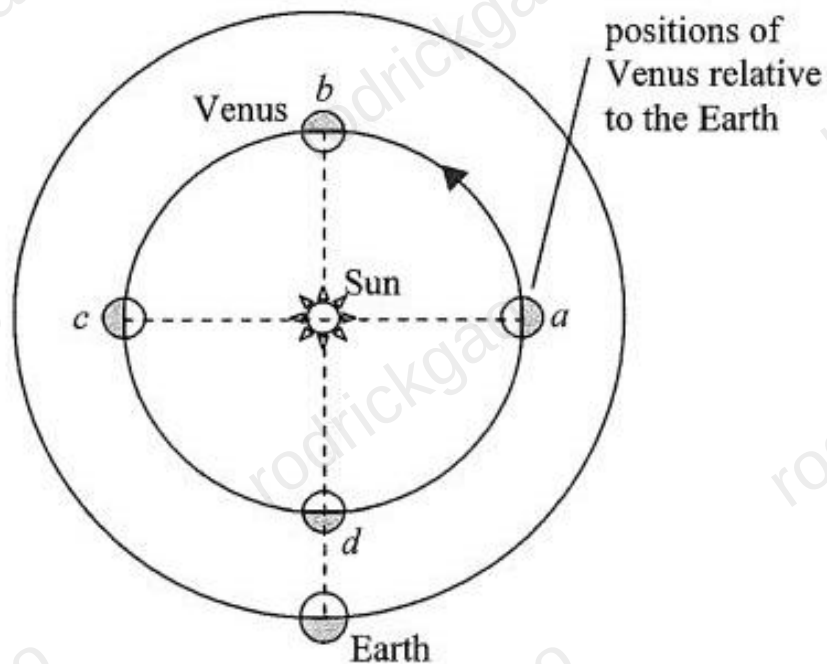
A. Both lines have the same fractional Doppler shift. $\lambda'/\lambda = (410 - 50)/410 = 360/410$, so $\lambda'_{\text{red}} = 656 \times 360/410 = 576$ nm. The absolute shift is not the same for different wavelengths.

Question 1.5

The following diagram is Galileo's drawing in 1610 showing the phases of Venus.



Which of the following parts of the orbit of Venus represent the above change of phases from right to left?



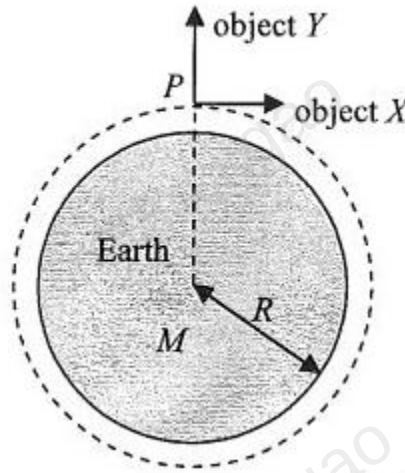
- A. $a \rightarrow b \rightarrow c$
- B. $b \rightarrow c \rightarrow d$
- C. $c \rightarrow d \rightarrow a$
- D. $d \rightarrow a \rightarrow b$

Official answer: B

B. At b, Venus is beyond the Sun and its illuminated face is almost fully visible, with a small angular size. Moving $b \rightarrow c \rightarrow d$ brings it closer to Earth, increasing angular size while the visible illuminated fraction decreases to a crescent.

Question 1.6

At a point P close to the Earth, two objects X, Y travel with the same speed $v = \sqrt{GM/R}$, with M and R being the mass and radius of the Earth respectively and G the universal gravitational constant. X travels tangentially at P while Y travels radially outward from P. Which of the following statements about their subsequent motions is correct? Neglect air resistance.



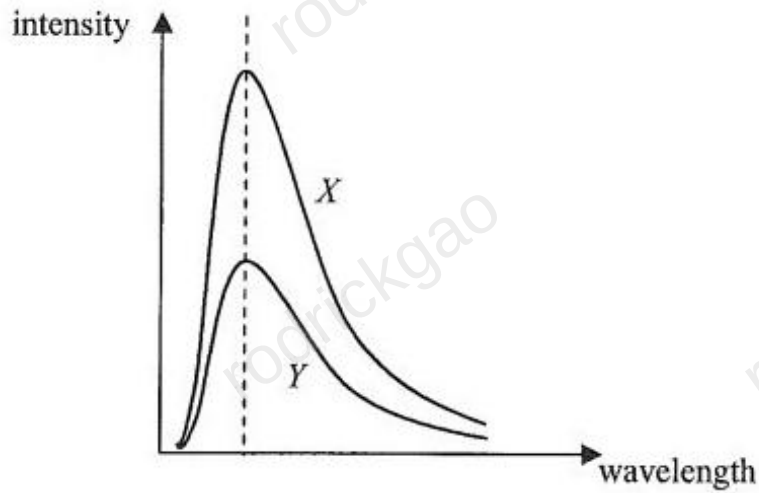
- A. X will eventually return to Earth; Y will continue to fly in its orbit.
- B. X and Y will both eventually return to Earth.
- C. X and Y will both continue to fly in their orbits.
- D. X will continue to fly in its orbit; Y will eventually return to Earth.

Official answer: D

D. $GMm/R^2 = mv^2/R$ gives exactly the stated speed for a tangential circular orbit. Y has no tangential speed and v is below escape speed $\sqrt{2GM/R}$; it rises, stops and falls back. Energy conservation gives its maximum distance from Earth's centre as $2R$.

Question 1.7

The diagram shows the spectra of radiation from stars X and Y with their peaks lying at the same wavelength.



Which statement is correct?

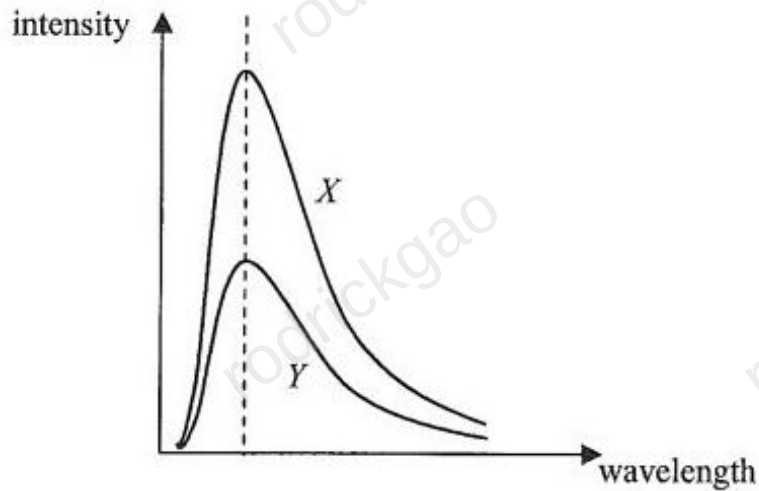
- A. Surface temperature of X > surface temperature of Y.
- B. Surface temperature of X < surface temperature of Y.
- C. Surface temperature of X = surface temperature of Y.
- D. The information is not sufficient to make a comparison of the surface temperatures of X and Y.

Official answer: C

C. Wien's displacement law $\lambda_{\text{max}}T = \text{constant}$ gives equal surface temperatures because the peak wavelengths coincide. The peak height does not determine temperature independently of what intensity represents.

Question 1.8

The diagram shows the spectra of radiation from stars X and Y with their peaks lying at the same wavelength.



Which statement is correct?

- A. Star X is smaller than star Y.
- B. Star X is bigger than star Y.**
- C. Star X and star Y are of the same size.
- D. The information is not sufficient to make a comparison of the sizes of stars X and Y.

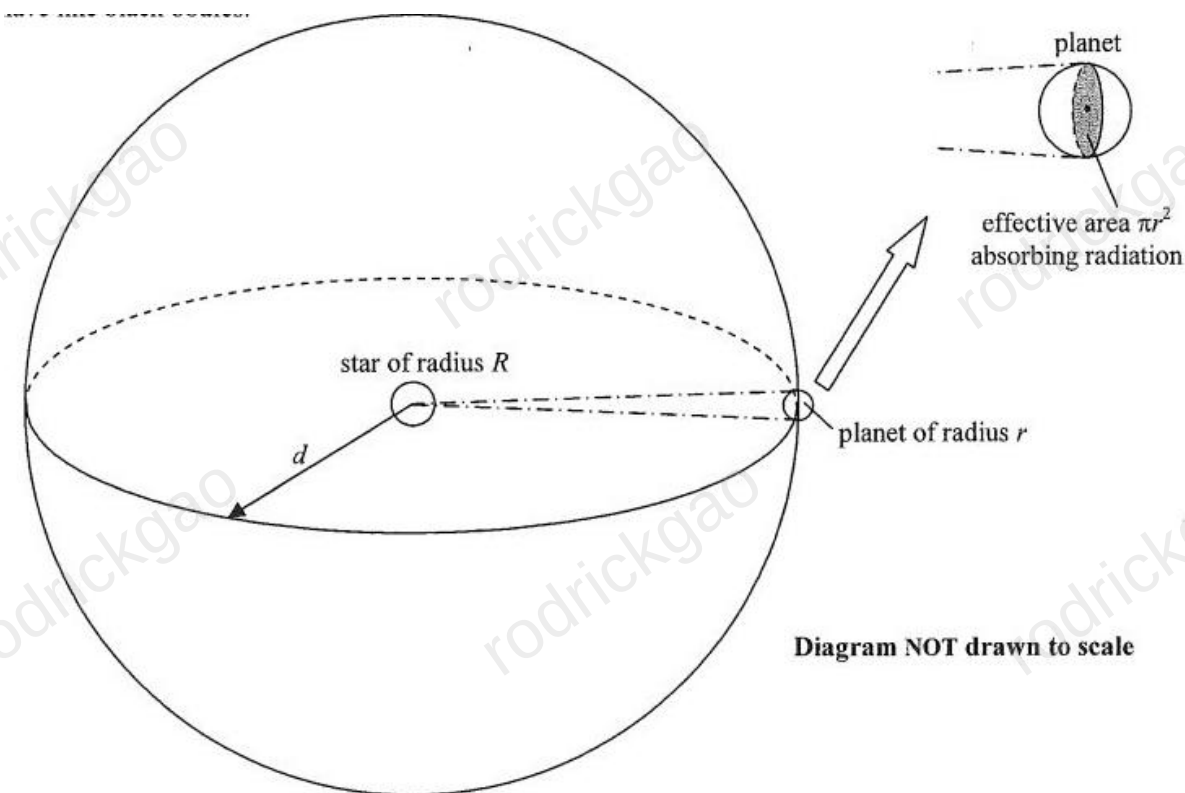
Official answer: B

B is the official key. Under the intended interpretation that the curves compare total emitted spectral power (or flux at equal distances), X has greater luminosity. Since T is equal and $L = 4\pi R^2 \sigma T^4$, $R_X > R_Y$.

Source ambiguity: the vertical axis says only “intensity”. If it means received flux at unspecified distances, $F = L/(4\pi d^2)$ does not determine the radii, and D would follow. The official B is retained with this assumption explicitly stated.

Question 1 Structured Question

(a) A star of radius R and surface temperature T_s (in K) emits radiation in all directions. A planet of radius r orbits the star at a distance d , which is much larger than both R and r . Assume that both the star and the planet behave like black bodies.



(a)(i) Taking the effective area that the planet absorbs radiation emitted from the star as πr^2 , show that the power absorbed by the planet is $\pi \sigma (rR/d)^2 T_s^4$, where σ is the Stefan constant. Assume that the planet is a perfect absorber of radiation. (2 marks)

(a)(ii) If the planet only absorbed energy, its temperature would rise indefinitely. However, this would not happen because the planet also radiates energy as it absorbs energy so that an equilibrium state is maintained. Show that the equilibrium surface temperature is $T_p = \sqrt{(R/(2d))} T_s$. (2 marks)

(b) A planet called Kepler-22b was discovered orbiting a Sun-like star with orbital radius 0.84 AU (1 AU = 1.50×10^{11} m). The star has radius 6.82×10^8 m and surface temperature 5518 K.

(b)(i) Estimate the equilibrium surface temperature of Kepler-22b using the results of (a). (2 marks)

(b)(ii) Liquid water is believed to be essential for life to exist on a planet. Based on (b)(i), explain whether Kepler-22b would be a favourable planet for life to exist or not. (2 marks)

(b)(iii) If Kepler-22b orbits a class K star instead of a Sun-like star (a class G star) with the same orbital radius, would its equilibrium surface temperature increase, decrease or remain unchanged? State your reason. Given: the sequence of spectral classes is O B A F G K M. (2 marks)

Official marking scheme

Solution		Marks	Remarks
1. (a)	(i) Luminosity of the star, $L = 4\pi R^2 \sigma T_s^4$ Power per unit area at distance d from the star $= \frac{L}{4\pi d^2} = \frac{R^2}{d^2} \sigma T_s^4$ Power absorbed $= \pi r^2 \times \frac{R^2}{d^2} \sigma T_s^4$	1M 1M 2	
	(ii) At equilibrium, power absorbed = power radiated $\frac{R^2}{d^2} \pi r^2 \sigma T_s^4 = 4\pi r^2 \sigma T_p^4$ $\frac{R^2}{d^2} T_s^4 = 4 T_p^4$ $T_p^4 = \frac{R^2}{4d^2} T_s^4$ $T_p = \sqrt{\frac{R}{2d}} T_s$	1M  1+1 1M 2	
	(b) (i) $T_p = \sqrt{\frac{R}{2d}} T_s$ $= \sqrt{\frac{6.82 \times 10^8}{2 \times (0.84 \times 1.50 \times 10^{11})}} \times 5518$ $= 287 \text{ K (or } 14^\circ \text{C)}$	1M 1A 2	
	(ii) The temperature is between 273 K and 373 K, (liquid) water is likely to exist on the planet. Hence the condition is favourable for life to exist.	1A 1A 2	
	(iii) The equilibrium surface temperature is lower / decreases. A class K star is a cooler star than a class G star.	1A 1A 2	

(a)(i) Spread the star's luminosity over a sphere of area $4\pi d^2$, then multiply the flux by the planet's projected disc πr^2 . This gives $\pi \sigma R^2 T_s^4 / d^2$.

(a)(ii) The planet intercepts radiation over a disc but emits over its whole spherical surface, $4\pi r^2$. Equating these powers and taking the positive fourth root gives $T_p = T_s \sqrt[4]{R/(2d)}$; r cancels.

(b)(i) $d = 1.26 \times 10^{11} \text{ m}$, giving $287.06 \text{ K} \approx 287 \text{ K} \approx 14^\circ \text{C}$. Use kelvin in the radiation law.

(b)(ii) This model temperature permits liquid water at an appropriate pressure, so the intended conclusion is favourable conditions, not proof of life. Atmospheric pressure and greenhouse effects are not supplied.

(b)(iii) The intended main-sequence comparison gives a cooler K star and a lower planetary temperature. The Chinese question explicitly says main sequence; the English question omits this qualifier. In general T_p also depends on $\sqrt{R}$, so spectral class alone without a radius or main-sequence assumption cannot establish the change.

Section B Atomic World

Question 2.1

Which of the following statements about Rutherford's atomic model is/are correct?

- (1) All the positive charges and nearly all of the atom's mass are concentrated in its tiny central nucleus.
- (2) Negatively charged electrons orbit around the nucleus.
- (3) The electrons can only move in certain orbits each having a definite energy and they do not radiate while accelerating.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1) and (2) and (3)

Official answer: A

A. Rutherford's model has a compact, massive positive nucleus with orbiting electrons. Quantized non-radiating stationary orbits are Bohr's additional postulate, so (3) does not belong to Rutherford's model.

Question 2.2

There are dark lines in the spectrum of sunlight. Which of the following statements are correct?

- (1) They are due to the absorption of certain wavelengths of light by the atoms in the Sun's atmosphere.
- (2) Light absorbed by the atoms in the Sun's atmosphere is then re-emitted in all directions.
- (3) The kinds of atoms present in the Sun's atmosphere can be deduced by the characteristics of the dark lines.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1) and (2) and (3)

Official answer: D

D. Atoms absorb photons whose energies match their energy gaps. Re-emission is in all directions, reducing the intensity along the original line of sight at those wavelengths. The characteristic set of wavelengths identifies the elements; all three statements are correct.

Question 2.3

A hydrogen atom in the ground state absorbs a photon of wavelength λ so that it is excited from quantum number $n = 1$ to $n = 3$. The ionization energy, in eV, of a hydrogen atom in the ground state is given by

A. $3hc/(2\lambda)$

B. $2hc/(3\lambda)$

C. $9hc/(8\lambda)$

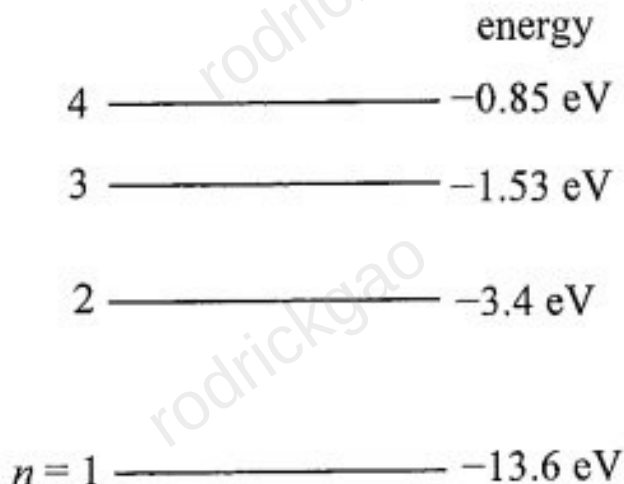
D. $8hc/(9\lambda)$

Official answer: C

C. Write $E_n = -E_1/n^2$. Then $hc/\lambda = E_3 - E_1 = (8/9)E_1$, hence $E_1 = 9hc/(8\lambda)$.

Unit issue in the printed source: with h in J s, this expression is in joules, and its numerical value in eV is $9hc/(8\lambda e)$. The printed options omit e ; C is correct when h is expressed in eV s. Both language editions and the official C are retained.

Question 2.4



The lowest four energy levels of a hydrogen atom are shown above. Which statement about a hydrogen atom in the ground state is **INCORRECT**?

- A. The collision between an electron of kinetic energy 10 eV and the atom is perfectly elastic.
- B. The collision between an electron of kinetic energy 12 eV and the atom can be inelastic.
- C. A photon of energy 12 eV can excite the atom to the first excited state.
- D. A photon of energy 14 eV can ionize the atom.

Official answer: C

C. The first excitation gap is $-3.4 - (-13.6) = 10.2$ eV. A photon must supply exactly this gap, so a 12 eV photon cannot produce that bound-state transition. A 12 eV electron can lose 10.2 eV and retain 1.8 eV; a 10 eV electron cannot excite the atom. A 14 eV photon exceeds the 13.6 eV ionization threshold.

The diagram's $n = 3$ label is transcribed as printed, -1.53 eV; ideal $E_n = -13.6/n^2$ gives about -1.51 eV. This discrepancy does not change the selected answer.

Question 2.5

When photons each of energy 3.41 eV are incident on a metal surface, the maximum kinetic energy of the photoelectrons emitted is 0.54 eV. What is the threshold frequency of this metal?

- A. 4.33×10^{33} Hz
- B. 9.53×10^{14} Hz
- C. 8.23×10^{14} Hz
- D. 6.93×10^{14} Hz

Official answer: D

D. Work function $\phi = E_{\text{photon}} - K_{\text{max}} = 3.41 - 0.54 = 2.87$ eV. Therefore $f_0 = \phi/h = (2.87 \times 1.60 \times 10^{-19}) / (6.63 \times 10^{-34}) = 6.93 \times 10^{14}$ Hz.

Question 2.6

A beam of light of frequency f falls on the cathode of a photocell so that photoelectrons are emitted. If the light beam is replaced by another one with the same intensity but having a frequency of $2f$, how would each of the following physical quantities change? Assume that each incident photon can emit one photoelectron.

V_s : stopping potential; I : magnitude of the saturation photoelectric current.

A. V_s increases; I increases.

B. V_s increases; I decreases.

C. V_s remains unchanged; I decreases.

D. V_s decreases; I increases.

Official answer: B

B. $eV_s = hf - \phi$, so doubling f increases V_s . At fixed incident power, doubling each photon's energy halves the photon arrival rate. With one electron per photon, the saturation current halves.

Question 2.7

The de Broglie wavelength of object X is shorter than that of object Y. Which of the following deductions must be correct?

(1) X has a higher speed than Y.

(2) X has a greater momentum than Y.

(3) X has greater kinetic energy than Y.

A. (2) only

B. (1) and (2) only

C. (2) and (3) only

D. (1) and (2) and (3)

Official answer: A

A. $\lambda = h/p$ implies $p_X > p_Y$. The masses are unspecified, so $v = p/m$ and $K = p^2/(2m)$ cannot be ordered from momentum alone. Only (2) must follow.

Question 2.8

Which statements about nanotechnology are correct?

(1) Glass can be made self-cleaning by coating it with titanium oxide in nanoscale.

(2) Carbon nanotubes and diamonds have the same spatial arrangement of carbon atoms.

(3) When gold is reduced to nanoscale, its melting point would be different from that of its bulk form.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1) and (2) and (3)

Official answer: B

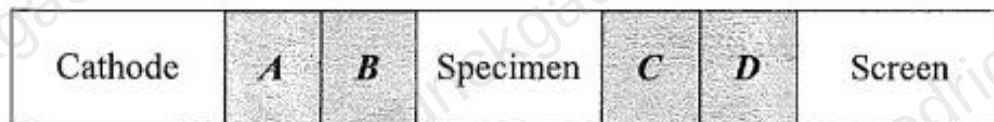
B. Nanoscale titanium dioxide provides photocatalytic self-cleaning. Carbon nanotubes have rolled graphitic networks, unlike diamond's three-dimensional tetrahedral network, so (2) is false. A large fraction of surface atoms changes nanoscale gold's melting behaviour.

Question 2 Structured Question

(a) In a Transmission Electron Microscope (TEM), electrons emitted from the cathode pass through the specimen and the four functional parts listed below before forming an image on a screen.

Functional parts: (1) objective magnetic lens; (2) projection magnetic lens; (3) condensing magnetic lens; (4) anode.

Referring to the following block diagram of a TEM, match the functional parts represented by A, B, C and D in the diagram. (2 marks)



(b)(i) When an electron of mass m and charge e is accelerated from rest by a voltage V , show that its de Broglie wavelength λ is given by $\lambda = h/\sqrt{2meV}$, where h is the Planck constant. (2 marks)

(b)(ii) The accelerating voltage of a TEM is 10 kV. Find λ . (2 marks)

(b)(iii) Explain why the resolving power of a TEM is higher compared with an optical microscope. (2 marks)

(c) Both Scanning Tunnelling Microscopes (STM) and Transmission Electron Microscopes (TEM) have very high resolving powers. Now if the internal structure of a slice of metallic specimen is to be studied, which of the above microscopes would be suitable or are both suitable? Explain. (2 marks)

Official marking scheme

Solution	Marks	Remarks
2. (a) A - (4): anode B - (3): condensing magnetic lens C - (1): objective magnetic lens D - (2): projection magnetic lens	2A 2	
(b) (i) K.E. = energy gain of the electron $\frac{1}{2}mv^2 = eV$ $(mv)^2 = 2meV$ $p = mv = \sqrt{2meV}$ $\therefore \lambda = \frac{h}{\sqrt{2meV}}$	1M 1M 2	
(ii) $\lambda = \frac{h}{\sqrt{2meV}}$ $= \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(10 \times 10^3)}}$ $\lambda = 1.2279 \times 10^{-11} \text{ m } (= 0.012 \text{ nm})$	1M 1A 2	$1.20 \times 10^{-11} \sim 1.23 \times 10^{-11} \text{ m}$
(iii) Since the wavelength of the electron beam ($\sim 10^{-11} \text{ m}$) is smaller / shorter than that of visible light ($\sim 10^{-7} \text{ m}$), resolving power of a microscope, $\theta = \frac{1.22\lambda}{d}$, is greater with shorter wavelength (or less diffraction with shorter wavelength).	1A 1A 2	
(c) TEM. STM only reveals surface structure of specimen.	1A 1A 2	

(a) The anode first accelerates the emitted electrons. The condenser focuses them on the specimen. After transmission, the objective forms an image and the projector enlarges it on the screen. Hence A, B, C, D = 4, 3, 1, 2.

(b)(i) Electrical work eV becomes kinetic energy. Using the non-relativistic relation $K = p^2/(2m)$ gives $p = \sqrt{2meV}$, then apply $\lambda = h/p$. Here e is the positive magnitude of electron charge.

(b)(ii) Convert 10 kV to 10^4 V before substituting: $\lambda = 1.2279 \times 10^{-11} \text{ m} = 0.01228 \text{ nm}$. This uses the non-relativistic model requested in (i).

(b)(iii) A shorter wavelength gives a smaller minimum resolvable angle $\theta_{\min} = 1.22\lambda/d$, so finer detail is distinguishable. The scheme calls this “greater resolving power”; it means better resolution, not a numerically larger θ .

(c) TEM transmits electrons through a thin specimen, revealing internal structure. STM measures tunnelling current between a tip and the conducting surface, so its image is surface information only.

Section C Energy and Use of Energy

Question 3.1

For monochromatic light sources, which of the following pairs appear(s) to be equally bright to our eyes?

- (1) 1 lumen of red light and 1 lumen of green light
- (2) red light of 1 watt and green light of 1 watt
- (3) red light of 1 watt and 1 lumen of red light

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

A. Luminous flux in lumens already weights radiant power by the eye's wavelength sensitivity. Equal lumens correspond to equal perceived brightness under the same viewing conditions. Equal watts of red and green do not, because the eye is more sensitive to green. A watt is not a lumen, so (3) is not generally equal.

Question 3.2

The energy labels for air-conditioners show the annual energy consumption (based on 1200 hours of operation per year) and the cooling capacity. Which air-conditioner below is the most energy efficient?

- A. Annual energy consumption 672 kW h; cooling capacity 2.44 kW.
- B. Annual energy consumption 684 kW h; cooling capacity 2.58 kW.
- C. Annual energy consumption 696 kW h; cooling capacity 2.89 kW.
- D. Annual energy consumption 750 kW h; cooling capacity 2.63 kW.

Official answer: C

C. Compare cooling output/input power, $\text{COP} = \text{cooling capacity}/(\text{annual consumption}/1200 \text{ h})$. The four values are 4.36, 4.53, 4.98 and 4.21. C supplies the most cooling per unit electrical energy, although its annual consumption is not the smallest.

Question 3.3

Which of the following building materials with thickness listed below give the best heat insulation?

- A. Concrete: thermal conductivity $0.50 \text{ W m}^{-1} \text{ K}^{-1}$; thickness 0.20 m.
- B. Wood: thermal conductivity $0.15 \text{ W m}^{-1} \text{ K}^{-1}$; thickness 0.05 m.
- C. Glass: thermal conductivity $1.00 \text{ W m}^{-1} \text{ K}^{-1}$; thickness 0.04 m.
- D. Plaster: thermal conductivity $0.24 \text{ W m}^{-1} \text{ K}^{-1}$; thickness 0.10 m.

Official answer: D

D. For equal area and temperature difference, heat transfer rate is $kA\Delta T/d$. Compare k/d : A 2.50, B 3.00, C 25.0, D $2.40 \text{ W m}^{-2} \text{ K}^{-1}$. D has the smallest heat flow, so it insulates best.

Question 3.4

Glass with low-emissivity coating can reduce heat transfer into a building by reflecting mainly

- A. ultra-violet radiation.
- B. visible light.
- C. infra-red radiation.
- D. microwaves.

Official answer: C

C. Low-emissivity glazing reflects thermal infrared radiation while allowing much visible light through, reducing radiative heat transfer. “Low emissivity” refers to thermal radiation, not nuclear radioactivity.

Question 3.5

A wind turbine generator experiences wind blowing normal to it with variable speed such that the wind speed is 1 m s^{-1} for the first two minutes and 2 m s^{-1} for the third minute. What is its average power output, in W, for this period of 3 minutes if the overall efficiency of the generator is 30% and the length of each blade is 20 m? Given: ρ = density of air in kg m^{-3} .

- A. $180\pi\rho$
- B. $200\pi\rho$
- C. $600\pi\rho$
- D. $667\pi\rho$

Official answer: B

B. Wind power through swept area A is $\frac{1}{2}\rho A v^3$. $A = \pi(20)^2 = 400\pi \text{ m}^2$. Electrical power = $0.30 \times \frac{1}{2}\rho \times 400\pi \times v^3 = 60\pi\rho v^3$. Time-weighted average = $[2(60\pi\rho) + 1(480\pi\rho)]/3 = 200\pi\rho \text{ W}$. Average v^3 , not v , because power is cubic in speed.

Question 3.6

Which of the following statements about hybrid vehicles is/are correct?

- (1) The battery of a hybrid vehicle needs to be recharged by an external electric source before the vehicle can run.
- (2) The power of the internal combustion engine of a hybrid vehicle is smaller than that of a conventional petrol vehicle of the same weight and performance.
- (3) The primary energy source of a hybrid vehicle is 100% petrol.

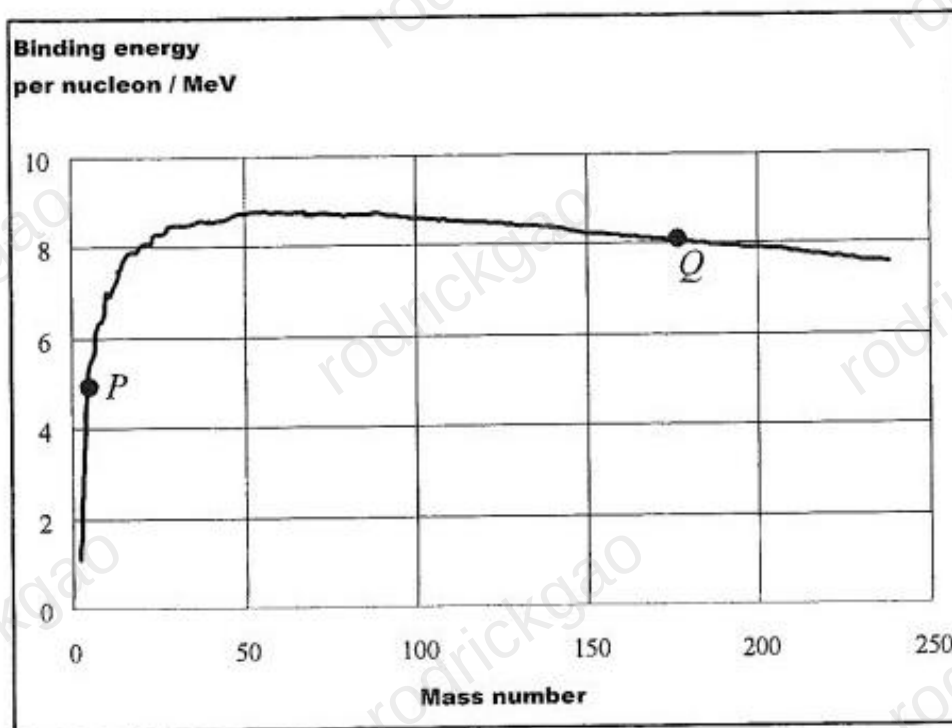
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: D

D. The question uses a conventional, non-plug-in petrol hybrid. Its engine and regenerative braking recharge the battery, so an external supply is unnecessary. Electric motor assistance permits a smaller combustion engine for the same performance. In this model, the original supplied energy is petrol; the battery is an intermediate store. Statements (2) and (3) are intended, not a claim about every type of hybrid.

Question 3.7

The curve below shows the binding energy per nucleon of nuclides of different mass numbers.



Which of the following statements is/are correct?

- (1) Nuclei of P can release energy by nuclear fusion.
- (2) Nuclei of Q can release energy by nuclear fission.
- (3) Nuclei of P are more stable than nuclei of Q.

A. (2) only

B. (1) and (2) only

C. (1) and (3) only

D. (1) and (2) and (3)

Official answer: B

B. Fusion of light P and fission of heavy Q can move the products toward the higher binding energy per nucleon near the middle of the curve. The increase in total binding energy is released. P has about 5 MeV per nucleon, less than Q's about 8 MeV, so P is less tightly bound, contradicting (3).

Question 3.8

Under normal operation, which of the following statements about a pressurized water reactor (PWR) of a nuclear power plant is/are correct?

- (1) The coolant which carries energy away from the reactor is radioactive.
- (2) The steam that drives the turbine is radioactive.
- (3) The cooling water discharged into the sea from the nuclear power plant contains some radioactive substances of the reactor.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Official answer: A

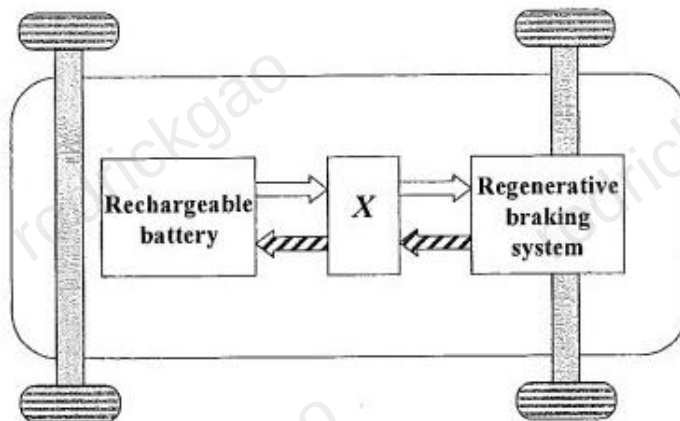
A. Primary coolant passes through the reactor and can become radioactive. It transfers heat through a heat exchanger to a separate secondary steam circuit; the seawater cooling circuit is separate again. During normal operation these fluids do not mix, so neither the turbine steam nor discharged cooling water contains reactor radioactivity in the stated model.

Question 3 Structured Question

(a) A completely discharged battery of an electric vehicle is fully charged to store 23 kW h of energy with a terminal voltage of 220 V at an average current of 13 A. Estimate the time in hours required to fully charge the battery. Neglect the internal resistance of the battery. (2 marks)

(b) Figure 3.1 shows the schematic diagram of an electric vehicle.

Figure 3.1



Open arrows denote transmission of energy when the vehicle is running; hatched arrows denote transmission of energy during braking.

(b)(i) What is the function of component X in Figure 3.1 when the vehicle is accelerating forward? Referring to Figure 3.1, describe how the regenerative braking system saves energy during braking. (3 marks)

(b)(ii) Assuming that a fixed percentage of energy is dissipated into heat during braking, would the regenerative braking system be more effective when the electric vehicle is moving at a low speed or a high speed? Explain. (2 marks)

(b)(iii) Why is it necessary for an electric vehicle also to be equipped with a mechanical braking system in addition to a regenerative braking system? (1 mark)

(c) Given that typical electric vehicles convert 60% of the electrical energy supplied into the vehicle's mechanical output, consider the following modes of operation:

Mode	Operation
1	Conventional petrol vehicles: 20% of energy stored in petrol is converted to the vehicle's mechanical output.
2	Coal-fired power plants + electric vehicles: coal-fired power plants are 45% efficient in converting energy stored in coal to electrical energy delivered at socket.
3	Nuclear power plants + electric vehicles: nuclear power plants are 35% efficient in converting energy stored in fuel rods to electrical energy delivered at socket.

Which mode has the highest **overall energy efficiency**? Does this mode have the **minimum overall emission of air pollutants** among the three modes? Explain your answer. (2 marks)

Official marking scheme

Solution	Marks	Remarks
3. (a) $E = VIt$ $23 \times 1000 = 220 \times 13 \times t$ $t = 8.04$ (hours)	1M 1A 2	28951 s or 482.5 minutes
(b) (i) Converting electrical / energy from battery to KE / mechanical energy / force to drive the car / accelerate the car <u>or</u> Motor During braking, some of the kinetic energy of the wheels / vehicle is converted by the motor / generator / component X to electrical energy. The electrical energy is then stored in / used to charge the rechargeable battery.	1A 1A 1A 3	
(ii) High speed. When braking at high speed, the amount of kinetic energy that can be converted to electrical energy (to recharge the battery) is larger.	1A 1A 2	KE of the car = E ; Energy dissipated as heat = αE Energy to recharge battery = $(1 - \alpha)E$ Fixed α . High speed $\Rightarrow$ More energy $(1 - \alpha)E$ to recharge battery. Accept same effect for high speed and low speed only if explanation has the meaning $\frac{E - \alpha E}{E} = (1 - \alpha)$
(iii) The mechanical braking system may come into play when the regenerative braking system fails. <u>Or</u> The mechanical braking system can lock the vehicle in position. <u>Or</u> The regenerative braking system will not work when the battery is fully charged.	1A 1A 1A 1	
(c) Mode 2 (overall efficiency = $45\% \times 60\% = 27\% > 20\%$ or 21% of the other two modes) No. Mode 3 practically has little or no emission of air pollutants.	1A 1A 2	

(a) Charging power = $VI = 220 \times 13 = 2860 \text{ W} = 2.86 \text{ kW}$. Time = energy/power = $23 \text{ kW h} / 2.86 \text{ kW} = 8.04196 \text{ h}$. Keeping energy in kW h avoids an unnecessary seconds conversion.

(b)(i) During acceleration X acts as a motor: battery electrical energy becomes vehicle kinetic energy. During braking it acts as a generator: the wheels drive it and kinetic energy becomes electrical energy stored chemically in the battery. Recovering this energy reduces the amount lost as heat.

(b)(ii) For the same mass, recoverable energy = $(1 - \alpha)\frac{1}{2}mv^2$. Thus higher initial speed allows more energy recovery, while the fraction recovered stays $1 - \alpha$. The scheme explicitly accepts the latter interpretation when explained.

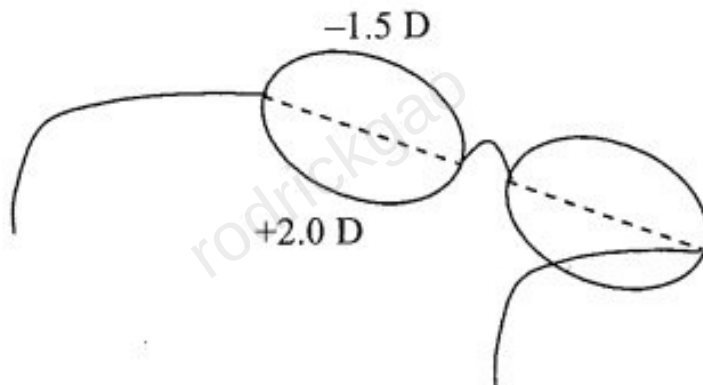
(b)(iii) A friction brake can hold the vehicle at rest and provide braking when the electrical system is unavailable or the full battery cannot accept charge. One valid reason earns the mark.

(c) Multiply efficiencies along each energy-conversion chain: $\eta_1 = 0.20$, $\eta_2 = 0.45 \times 0.60 = 0.27$, $\eta_3 = 0.35 \times 0.60 = 0.21$. The most efficient chain need not be the cleanest: coal combustion emits air pollutants, whereas the nuclear mode has little direct operational air-pollutant emission.

Section D Medical Physics

Question 4.1

Mr. Lee wears a pair of bifocal lenses as shown. The respective powers of the upper half and the lower half of each lens are -1.5 D and $+2.0\text{ D}$. Which of the following statements is/are correct?



- (1) The upper half is for viewing distant objects while the lower half is for viewing objects at a close distance.
- (2) Mr. Lee only suffers from old sight (presbyopia).
- (3) Without the spectacles, Mr. Lee cannot see an object clearly no matter how far it is placed from him.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Official answer: A

A. The upper diverging lens corrects myopia for distance viewing; the lower converging portion assists near viewing when accommodation is inadequate. He therefore also has myopia, not only presbyopia. The -1.5 D distance correction corresponds to an unaided far point about $1/1.5 = 0.67\text{ m}$, where he can see clearly, so (3) is false.

Question 4.2

Which of the following statements about the threshold of hearing is/are correct?

- (1) The intensity of sound of the threshold of hearing is 0 W m^{-2} .
- (2) The corresponding sound intensity level of the threshold of hearing is chosen as 0 dB .
- (3) The threshold of hearing depends on the frequency of sound.

A. (1) only

B. (2) only

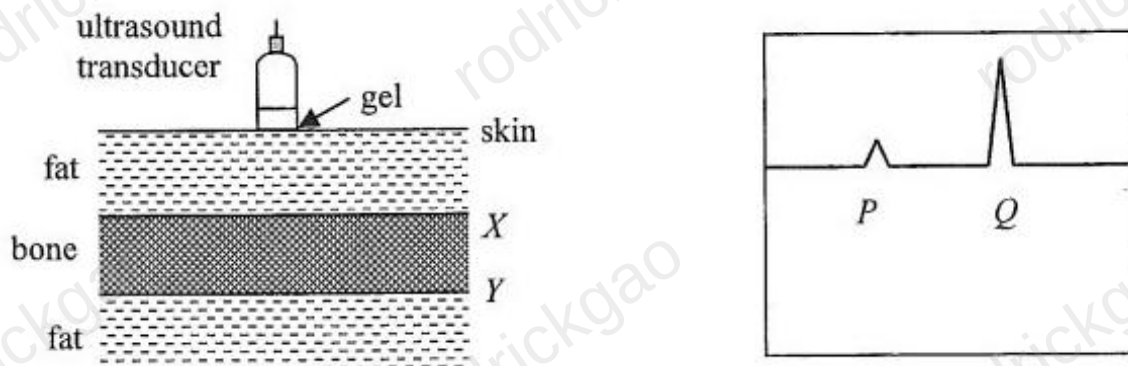
C. (1) and (3) only

D. (2) and (3) only

Official answer: D

D. The conventional reference intensity $I_0 = 10^{-12}\text{ W m}^{-2}$ is nonzero; $10 \log_{10}(I_0/I_0) = 0\text{ dB}$. Actual hearing sensitivity depends on frequency. Thus (2) is the reference convention used in the question and (3) is correct; 0 dB does not mean zero sound energy.

Question 4.3



An ultrasound transducer is placed on the skin over a certain position of the human body to perform an A-scan. The signal received contains two spikes P and Q as shown. Which of the following statements is/are correct?

- (1) There is almost no reflection from interface Y because the bone absorbs nearly all of the ultrasound.
- (2) There is almost no reflection from interface Y because interface X reflects nearly all of the ultrasound.
- (3) The spikes P and Q correspond to the reflections at interfaces X and Y respectively.

- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

Official answer: B

B. The intended schematic treats the fat–bone interface X as a strong reflector because of the acoustic-impedance mismatch. Little ultrasound reaches the deeper interface Y, so it gives no appreciable echo. Q is the strong echo at X, while P is an earlier shallow-interface echo; they are not X and Y respectively. The suppression is attributed to reflection at X, not near-total absorption within bone.

The words “nearly all” are the exam’s idealization; the reflected fraction in a quantitative model is $[(Z_{\text{bone}} - Z_{\text{fat}})/(Z_{\text{bone}} + Z_{\text{fat}})]^2$, not universally 100%.

Question 4.4

Which statement about fibre optic endoscopes is correct?

- A. A coherent fibre bundle is mainly used for transporting light from a light source to the area under examination and back to the other end.
- B. Non-coherent fibre bundles have a greater amount of light lost compared to coherent fibre bundles.
- C. Both coherent and non-coherent fibre bundles can function normally when bent slightly.
- D. Only black and white images can be displayed by fibre optic endoscopes.

Official answer: C

C. Total internal reflection continues through a gently bent fibre. Coherence means preserving the relative fibre positions at both ends to relay an image; illumination can use a non-coherent bundle. Coherence alone does not set transmission loss, and visible wavelengths can transmit colour images.

Question 4.5

A patient is going to take a needle aspiration biopsy in which a fine needle is inserted into his liver through the skin to take a tiny living tissue for testing. In order to minimize the risk of internal bleeding, it is important to locate the large blood vessels of the liver near the place where the needle is inserted. Also, as the liver can displace slightly inside the body, real-time imaging is therefore needed during needle insertion. The most suitable imaging method is

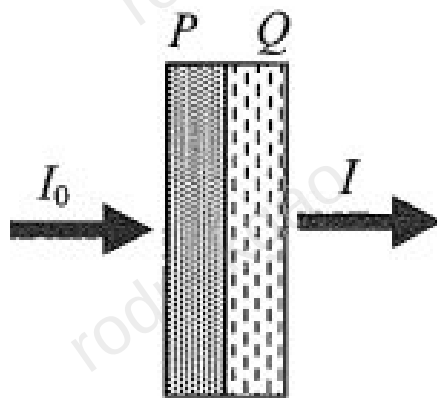
- A. X-ray planar imaging.
- B. computed tomography.
- C. ultrasound imaging.
- D. radionuclide imaging.

Official answer: C

C. For the scenario in the question, ultrasound can display soft tissue and vessel positions in real time while the needle is guided. It avoids ionizing radiation during continuous observation. Plain X-rays give poor soft-tissue contrast, and radionuclide imaging is not suited to locating a needle relative to nearby vessels.

Question 4.6

An object is made up of two different materials P and Q of 1 cm equal thickness as shown. The linear attenuation coefficients of P and Q for X-rays are 0.05 cm^{-1} and 0.68 cm^{-1} respectively. An X-ray beam of intensity I_0 is incident on the object and emerges from the object with an intensity I . Which of the following expressions gives the ratio I/I_0 ?



- A. $0.05/0.68$
- B. $(0.68 - 0.05)^2/(0.68 + 0.05)^2$
- C. $e^{-0.05/0.68}$
- D. $e^{-(0.05 + 0.68)}$

Official answer: D

D. Apply $I = I_0 e^{-\mu x}$ successively to both layers. The transmission factors multiply: $I/I_0 = e^{-0.05 \times 1} e^{-0.68 \times 1} = e^{-0.73} \approx 0.482$. The attenuation coefficients use cm^{-1} , so use each thickness as 1 cm.

Question 4.7

What is the function of artificial contrast medium (ACM) when taking radiographic images?

- A. It slows down the X-rays so that the X-rays stay longer in the organ.
- B. It increases the absorption of some of the X-rays by the organ.
- C. It increases the energy of X-rays before the X-rays leave the organ.
- D. After injection, the ACM in aqueous form decreases the density of the organ that is undergoing radiographic imaging.

Official answer: B

B. A contrast agent containing strongly attenuating elements increases X-ray absorption in the filled organ or vessel. The resulting difference in transmitted intensity makes it stand out from surrounding tissue; it does not work by slowing the X-rays or increasing their energy.

Question 4.8

The biological half-life and physical half-life of a radionuclide Y are two days and three days respectively. A patient takes in a quantity of Y as tracer for radionuclide imaging. How long does it take for the amount of radionuclide Y left in the patient's body to reduce to $1/8$ of the original intake?

- A. 3.6 days
- B. 4.8 days
- C. 7.5 days
- D. 15 days

Official answer: A

A. Independent biological removal and radioactive decay multiply their remaining fractions: $N/N_0 = 2^{-t/2}2^{-t/3} = 2^{-5t/6}$. Set this equal to $1/8 = 2^{-3}$: $5t/6 = 3$ and $t = 3.6$ days. Equivalently, $T_{\text{eff}} = (1/2 + 1/3)^{-1} = 1.2$ days, followed by three effective half-lives.

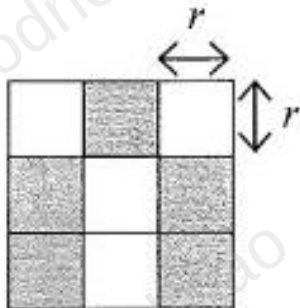
Question 4 Structured Question

(a)(i) In medical imaging using ultrasound, a piezoelectric transducer is employed to scan the patient. Describe how a piezoelectric transducer generates ultrasound waves. (2 marks)

(a)(ii) State ONE advantage and ONE disadvantage of using ultrasound of high frequencies in medical imaging. (2 marks)

(b)(i) John has normal eyesight and the power of his eye is +59 D in viewing distant objects. Estimate the separation between the lens and the retina of his eye. Assume that the refracting power is mainly contributed by the eye lens. (2 marks)

(b)(ii) The display panel of a smart phone X is made up of numerous square pixels as shown.



square pixels of part of the display panel

John is looking at the graphics on the display panel of smart phone X. The diameter of his eye's pupil is 4.0 mm. Estimate the resolving power θ (in radians) of his eye for graphics in green colour. Given: wavelength of green light = 5.35×10^{-7} m. (2 marks)

(b)(iii) The pixels of smart phone X are so small that the human eye is unable to distinguish two adjacent pixels at a typical viewing distance $L = 0.30$ m. Using (b)(ii), estimate the maximum length of a side of a square pixel, r , on the display panel of smart phone X. You may assume that for small angle θ in radians, $\tan \theta \approx \theta$. (2 marks)

Official marking scheme

Solution		Marks	Remarks
4.	(a) (i)	When applying a potential difference across a small block of piezoelectric crystal inside the transducer, the crystal will be distorted; it will return to its original shape if the potential difference is removed, thus ultrasound waves will be generated due to its subsequent oscillations.	1A
			1A
			2
	(ii)	Advantage : better resolution / clearer Disadvantage: greater attenuation / penetrate less	1A
			1A
			2
	(b) (i)	$P = \frac{1}{u} + \frac{1}{v}$ or $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $59 = \frac{1}{\infty} + \frac{1}{v}$ $v = 0.01695 \text{ m (or } 1.695 \text{ cm)} \approx 17 \text{ mm}$	1M
			1A
			Accept 16.9 ~ 17.0 mm
			2
	(ii)	$\theta = \frac{1.22 \lambda}{d}$ $\theta = \frac{1.22 \times 5.35 \times 10^{-7}}{4.0 \times 10^{-3}}$ $= 1.63175 \times 10^{-4} \text{ (rad)} \approx 1.63 \times 10^{-4} \text{ (rad)}$	1M
			1A
			Accept 0.0093°
	(iii)	$\theta = \frac{r}{L}$ for small θ in radians $r = 1.632 \times 10^{-4} \times 0.30 \text{ m}$ $= 4.89525 \times 10^{-5} \text{ m (or } 0.0489525 \text{ mm)} \approx 49.0 \text{ }\mu\text{m}$	1M
			1A
			Accept $\tan \theta = \frac{r}{L}$ Accept 48.9 ~ 49.0 μm
			2

(a)(i) A voltage changes the crystal's dimensions through the inverse piezoelectric effect. Removing or reversing the voltage lets it recover or deform oppositely. Rapid repeated electrical excitation makes it oscillate at an ultrasonic frequency, launching pressure waves into the coupled medium.

(a)(ii) At the same wave speed, $\lambda = c/f$ decreases as frequency rises, giving finer spatial resolution. Higher-frequency ultrasound is attenuated more strongly in tissue, so it penetrates less deeply.

(b)(i) A distant object has $u \approx \infty$. For a normal eye its image falls on the retina, so lens-retina separation $v = f = 1/P = 1/59 \text{ m} = 16.949 \text{ mm} \approx 17 \text{ mm}$.

(b)(ii) Use the pupil diameter, not its radius: $d = 4.0 \times 10^{-3} \text{ m}$. Rayleigh's circular-aperture criterion gives $\theta_{\min} = 1.63175 \times 10^{-4} \text{ rad}$. A smaller θ means better resolution.

(b)(iii) Adjacent pixel centres are separated by r , subtending angle r/L . The boundary of distinguishability is $r/L = \theta_{\min}$, giving $r_{\max} = 0.30 \times 1.63175 \times 10^{-4} = 4.89525 \times 10^{-5} \text{ m} = 48.9525 \text{ }\mu\text{m} \approx 49.0 \text{ }\mu\text{m}$. The scheme prints a rounded 1.632 in the intermediate line but retains the unrounded product.